Aisaiah Maguadog

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Education

Loyola Marymount University — Los Angeles, CA
Bachelor of Science in Computer Engineering — Expected May 2027
Relevant Coursework: Computer Architecture, Embedded Systems, Machine Learning, Data Structures & Algorithms, Signals & Systems

Technical Skills

Languages & HDL: SystemVerilog, C, Python, VHDL, Verilog, ARM Assembly, MIPS Assembly, C++
RTL & Verification: RTL Design, FSM Design, Datapath/Control Design, Structural Modeling, Testbench Development, RTL Simulation, Waveform Debugging, Fixed-Point Verification
Embedded Systems & SoC: Zynq SoC, AXI, AXI GPIO, Memory-Mapped I/O, Interrupt-Driven I/O, UART, SPI
Platforms & Tools: Vivado, Vivado IP Integrator, ZedBoard, Raspberry Pi, Jetson Orin Nano, Linux, ROS 2, Git, GitHub
Lab & Simulation: Oscilloscope, Logic Analyzer, Function Generator, Multimeter, Soldering, Multisim, LabVIEW

Projects

Formula SAE Distributed Data Acquisition & Telemetry System – Capstone — Fall 2026 – Present
- Architecting a distributed Formula SAE data-acquisition system, defining quantitative requirements for telemetry bandwidth, latency, timing accuracy, reliability, and subsystem interfaces
- Evaluated 10BASE-T1S with PLCA for deterministic multidrop vehicle networking; developed an IEEE 1588-based timing architecture for synchronized sensor acquisition across distributed nodes
- Designing fault-tolerant data handling with hardware listen-only CAN monitoring, ring-buffered telemetry storage, retransmission mechanisms, and an STM32U5-based driver dashboard

SWIFT Frequency-Tracking Algorithm – SystemVerilog RTL — Spring 2026
- Designed fixed-point SystemVerilog RTL for a SWIFT frequency-tracking architecture, including a Q1.15 × Q9.14 complex multiplier, 64-bin state RAM, and coefficient ROM
- Developed SystemVerilog testbenches to verify complex arithmetic, fixed-point behavior, and synchronous memory timing; used waveform analysis to debug module interfaces in preparation for top-level FSM integration

32-Bit Single-Cycle MIPS Processor – SystemVerilog — Spring 2026
- Designed and integrated a 32-bit single-cycle MIPS processor in SystemVerilog, developing the ALU, dual-read register file, instruction/data memories, control logic, and datapath as modular RTL components
- Implemented and verified R-type, `addi`, `lw`, `sw`, and `beq` instruction paths using module-level and top-level testbenches, RTL simulation, and waveform debugging

Zynq SoC Embedded I/O System – C / Vivado — Fall 2026
- Built Zynq SoC hardware/software systems in Vivado IP Integrator using the ARM Cortex-A9 processing system, programmable logic, AXI interconnect, and memory-mapped AXI GPIO peripherals
- Developed interrupt-driven C applications using push-button ISRs to drive binary counter output on LEDs; integrated UART for serial-terminal debugging
- Generated FPGA bitstreams, exported hardware platforms to Xilinx SDK, and validated designs on ZedBoard hardware

Discrete BJT Operational Amplifier – Analog Circuit Design — Spring 2026
- Designed and built a two-stage discrete BJT op-amp using CA3046 NPN and MPSA13 Darlington devices, achieving 400–500 V/V gain, $R_{in} \geq 20\,\text{k}\Omega$, $R_{out} \leq 200\,\Omega$, and $f_L \leq 100\,\text{Hz}$
- Applied Miller compensation and designed a 20 V/V inverting amplifier; characterized gain, bandwidth, CMRR, input/output impedance, and clipping behavior against Multisim simulation

Experience

Apple — Manhattan Beach, CA
Specialist — Aug 2021 – Present
- Diagnose hardware, software, networking, account, and ecosystem issues across iPhone, Mac, iPad, Apple Watch, and Apple Vision Pro, translating technical findings into customer-facing solutions
- Support consumer, education, and business customers across Apple hardware, operating systems, and services
- Coordinate across sales, technical support, operations, and leadership to resolve complex cases in a high-volume environment

Leadership & Awards

- **1st Place, NSBE Hackathon** – National Society of Black Engineers
- **Vice President, IEEE LMU Student Branch** – technical workshops, career events, chapter operations
- **Vice President, LMU Cube Satellite Club** – CubeSat activities and guest-speaker events
- **Vice President, ROBO** – robotics projects and club operations